Homework 03

Submission Notices:

- *Conduct your homework by filling answers into the placeholders given in this file (in Microsoft Word format). Questions are shown in black color, instructions/hints are shown in italic and blue color, and your content should use any color that is different from those.*
- *After completing your homework, prepare the file for submission by exporting the Word file (filled with answers) to a PDF file, whose filename follows the following format,*
  <StudentID-1>_<StudentID-2>_HW02.pdf (Student IDs are sorted in ascending order)
  *E.g., **2252001_2252002_HW02.pdf***
  and then submit the file to Moodle directly WITHOUT any kinds of compression (.zip, .rar, .tar, etc.).
- *Note that you will get zero credit for any careless mistake, including, but not limited to, the following things.*
  1. *Wrong file/filename format, e.g., not a pdf file, use "-" instead of "_" for separators, etc.*
  2. *Disorder format of problems and answers*
  3. *Conducted not in English.*
  4. *Cheating, i.e., copying other students' works or let the other student(s) copy your work.*

Problem 1. (3pts) Consider a first-order knowledge base that describes worlds containing people, songs, albums (e.g., "Meet the Beatles") and disks (i.e., particular physical instances of CDs). The vocabulary contains the following symbols:

- CopyOf(d, a): Disk d is a copy of album a
- Owns(p, d):Person p owns disk d
- Sings(p, s, a): Album a includes a recording of song s sung by person p
- Wrote(p, s): Person p wrote song s

Express the following statements in first-order logic:

1. Gershwin did not write "Eleanor Rigby."
2. Either Gershwin or McCartney wrote "The Man I Love."
3. Joe owns a copy of *Revolver*.
4. Every song that McCartney sings on *Revolver* was written by McCartney.
5. Gershwin did not write any of the songs on *Revolver*.
6. Every song that Gershwin wrote has been recorded on some album. (Possibly different songs are recorded on different albums.)
7. There is a single album that contains every song that Joe has written
8. Joe owns a copy of an album that has Billie Holiday singing "The Man I Love."
9. Joe owns a copy of every album that has a song sung by McCartney. (Of course, each different album is instantiated in a different physical CD.)

Please fill your answers in the table below.

No.	Original statements and FOL sentences
1.	Gershwin did not write "Eleanor Rigby."
	¬Wrote(Gershwin, Eleanor Rigby)
2.	Either Gershwin or McCartney wrote "The Man I Love."

	Wrote(Gershwin, The Man I Love) ∨ Wrote(McCartney, The Man I Love)
3.	Joe owns a copy of *Revolver*.
	∃d (Owns(Joe, d) ∧ CopyOf(d, Revolver))
4.	Every song that McCartney sings on *Revolver* was written by McCartney.
	∀s ((Sings(McCartney, s, a) ⟶ Wrote(McCartney, s))
5.	Gershwin did not write any of the songs on *Revolver*.
	$\forall s\big(\exists p\ Sings(p, s, Revolver)\big) \rightarrow\ \neg Wrote(Gershwin, s)$
6.	Every song that Gershwin wrote has been recorded on some album. (Possibly different songs are recorded on different albums.)
	$\forall s\ Wrote(Gershwin, s) \rightarrow \exists a, p\ Sings(p, s, a)$
7.	There is a single album that contains every song that Joe has written
	$\exists a \exists s\ Wrote(Joe, s) \wedge\ \forall s' \neg Wrote(Joe, s') \wedge\ s \neq s'\ \ \rightarrow \exists\ p\ Sings(p, s, a)$
8.	Joe owns a copy of an album that has Billie Holiday singing "The Man I Love."
	$\exists d, a\ CopyOf(d, a) \wedge Owns(Joe, d) \wedge Sings(Billie\ Holiday, The\ Man\ I\ Love, a)$
9.	Joe owns a copy of every album that has a song sung by McCartney. (Of course, each different album is instantiated in a different physical CD.)
	$\forall a\big(\exists s\ Sings(McCartney, s, a)\big) \rightarrow \exists d\ Owns(Joe, d) \wedge CopyOf(d, a)$

Problem 2. (2pts) Consider a vocabulary with the following symbols:

Occupation(p, o): Predicate. Person p has occupation o. [O(p, o)]
Customer (p1, p2): Predicate. Person p1 is a customer of person p2. [C(p1, p2)]
Boss(p1, p2): Predicate. Person p1 is a boss of person p2. [B(p1, p2)]
Doctor , Surgeon, Lawyer , Actor : Constants denoting occupations. [D, S, L, A]
Emily, Joe: Constants denoting people. [E, J]

Translate the following FOL sentences into English.

Please fill your answers in the table below.

No.	Original statements and FOL sentences
1.	O(E, S) ∨ O(E, L)
	Either Emily has an occupation as a surgeon or a lawyer.
2.	O(J, A) ∧ ∃p, p ≠ A ∧ O(J, p)
	Hoe has another job besides being an actor.
3.	∀p, O(p, S) → O(p, D)

	Every surgeon is a doctor.
4.	$\neg\exists p, C(J, p) \land O(p, L)$
	None of Joe's clients are lawyers.
5.	$\exists p, B(p, E) \land O(p, L)$
	Emily has a boss who is a lawyer.
6.	$\exists p, O(p, L) \land \forall q, C(q, p) \rightarrow O(q, D)$
	If one of the customers is a lawyer, then they have a profession as a doctor.
7.	$\forall p, O(p, S) \rightarrow \exists q, O(q, L) \land C(p, q)$
	Anyone who works as a surgeon also has a customer who is a lawyer.

Problem 3. (3pts) Consider the following statements. *Tony, Shi-Kuo and Ellen belong to the Hoofers Club. Every member of the Hoofers Club is either a skier or a mountain climber or both. No mountain climber likes rain, and all skiers like snow. Ellen dislikes whatever Tony likes and likes whatever Tony dislikes. Tony likes rain and snow.*

a) Let S(x) mean "x is a skier", M(x) mean "x is a mountain climber", and L(x, y) mean "x likes y", where the domain of the first variable is Hoofers Club members, and the domain of the second variable is snow and rain. Build a FOL knowledge base from the above statements, using only given predicates. (1pt)

Please fill your answers in the table below.

No.	Clauses
1.	$\forall x\,(S(x) \lor M(x))$
2.	$\forall x\,(M(x) \rightarrow \neg L(x, rain))$
3.	$\forall x\,(S(x) \rightarrow L(x, snow))$
4.	$\forall x\,(L(E, x) \leftrightarrow \neg L(T, x))$
5.	$L(T, rain)$
6.	$L(T, snow)$

b) Convert each FOL sentence in the knowledge base into CNF. (1pt)

Please fill your answers in the table below.

No.	Clauses
1.	$S(x) \lor M(x)$
2.	$\neg M(x) \lor \neg L(x, rain)$

3.	¬S(x) ∨ L(x, snow)
4.	¬L(E, x) ∨ ¬L(T, x)
5.	L(E, x) ∨ ¬L(T, x)
6.	L(T, rain)
7.	L(T, snow)

c) Use resolution refutation to answer the query, "*Is there a member of the Hoofers Club who is a mountain climber but not a skier?*" (1pt)
Please fill your answers in the table below.

No.	Clauses	From which sentences?
8.	¬M(x) ∨ S(x)	negated query
9.	S(x)	1 and 8
10.	L(x, snow)	3 and 9
11.	L(E, snow)	5 and 10
12.	¬L(T, snow)	4 and 11

Problem 4. (1pt) Given a chess board with 4 rows and 4 columns (4x4) as below.

			

Assign a Boolean variable to each cell of the board as below (*1, 2, 3, etc. are variable names*)

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

Write CNF clauses for each of the following cases when placing a queen in cell [6].
Please fill your answer in the table below.

No.	Clauses
1.	There is a queen in cell [6] if and only if there is no queen in the same row.
	$6 \wedge \neg 5 \wedge \neg 7 \wedge \neg 8$
2.	There is a queen in cell [6] if and only if there is no queen in the same column.
	$6 \wedge \neg 2 \wedge \neg 10 \wedge \neg 14$
3.	There is a queen in cell [6] if and only if there is no queen in the same major diagonal.
	$6 \wedge \neg 1 \wedge \neg 11 \wedge \neg 16$
4.	There is a queen in cell [6] if and only if there is no queen in the same minor diagonal.
	$6 \wedge \neg 3 \wedge \neg 9$

Problem 5. (1pt) Yi and Lily are two cave explorers. Once when they explored a new cave, they found 3 boxes and some instructions carved on the cliff as follows: *"Inside the 3 boxes below, only one box will be filled with gold, the remaining 2 boxes will not contain gold (but are poisonous). On their lids has imprinted on it a clue as to its contents."*

The clues are:

Box 1 "The gold is not here."

Box 2 "The gold is not here."

Box 3 "The gold is in Box 2."

Know that **"***Only one message is true; the other two are false. Which box has the gold?"*

Help Yi and Lily resolve their issue by formalizing the puzzle in Propositional Logic and find the solution using a truth table.

a) Let B_i ($i \in \{1,2,3\}$) stand for "gold is in the *i-th* box." Formalize the statements of the problem. (0.5pt).

Please fill your answer in the given table below.

Statement	**Convert to CNF**	
One box contains gold, the other two are empty.	$(B_1 \wedge \overline{B_2} \wedge \overline{B_3}) \vee (\overline{B_1} \wedge B_2 \wedge \overline{B_3}) \vee (\overline{B_1} \wedge \overline{B_2} \wedge B_3)$	**(1)**
Only one message is true; the other two are false.	B_1	**(2)**

b) Compute the below truth table. Add more rows as needed.

B_1	B_2	B_3	(1)	(2)
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
1	0	0	1	1
0	1	1	0	0
1	0	1	0	1
1	1	0	0	1
1	1	1	0	1

Conclusion: Box 1 has the goal.